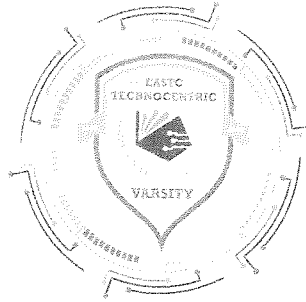


LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 05	Wireways and Wiring, NQF Level 4.	11



M. Ndabeni

36 / 36 = 100%
140
90%

WIRE WAYS AND WIRING

SUMMATIVE ASSESMENTS

Module #	671101000-KM-05:
NQF Level	level 4,
Notional hours	110
Credit(s)	Cr 11
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	Muyisi
Surname	Ndabeni
Cell NUMBER	083 800 4724
E-mail	mvndabeni@gmail.com

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed

Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Ndabeni*

Learners Name: *Munyisi Ndabeni*Learner's signature: *[Signature]*

Date: _____

Assessors Name: _____

Assessor's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments

- If not all evidence available was taken into account during the assessment
- The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
- The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
- Not all the range items were available for assessment

2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.



ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____



PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Munyer Ndabeni</i>	Signature <i>[Signature]</i>
Assessors Name:	Signature:



INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1: KM-05-KT01: Wire ways (35%)

Learning Outcome 1:	KT0101 Definition, types, purpose and applications of wireways
Learning Outcome 2:	KT0102 General provisions for wireways as prescribed in the SANS 10142-1

Task	Questions Description	Marks
1	a) Define the term wireway as given in the SANS 10142-1 Code of Practice	5
	b) Describe different types of wireways and their applications	5

Wire Way is the latest South African standard for low-voltage electrical installation (up to 1kV), focusing on safety, compliance, and modernized energy system. Key updates include arc fault detection, new earthing requirements, and solar PV integration, mandatory for all new, altered, or repaired installations.

Aspects of SANS 10142-1:2024 (ed. 3.2)

Latest edition: Edition 3.2 was released in August 2024, addressing previous amendments and updating standards.

Scope: It dictates requirements for wiring, distribution boards, switchgear, and safety procedures from the point of consumption.

Compliance: An updated installation requires a new certificate of compliance (COC) based on the latest standards.

Always ensure your electrician is using the most recent amendment of the code, as the standard is periodically updated for safety improvements.

(b) Lay-in wireways (Hinged cover): feature a hinged cover on a side that opens completely, allows cables to be laid in rather than pulled through, reducing installation time. Often used in industrial settings, control panels, and machinery wiring, including NEMA 12 types which are designed to protect against dust, oil, and water.

Pull-through wireways (standard trough): Require cables to be pulled through the entire length. These are often used for long, straight runs of conduit in commercial or industrial building sections.

Cable troughs (open cable trays): An open-bottom, ventilated structure used for supporting large quantities of power cables. They provide excellent ventilation to prevent overheating in data centers, industrial plants, and commercial buildings.

Non-metallic wireways: Made from PVC or other materials, these are resistant to corrosion and moisture. They are ideal for wet, corrosive environments or areas where metallic conduits would degrade.

LEARNING UNIT 2:

KM-05-KT02: Wiring of installations (35%)

Learning Outcome 1:	KT0201 Regulations and statutory requirements for wiring of premises
Learning Outcome 2:	KT0202 Electrical diagrams and symbols
Learning Outcome 3:	KT0203 Electrical components and their applications

Task	Questions Description	Marks
2	List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings	10

IEC 60617 (formerly BS 3939) is the international standard for electrical symbols, providing a standardized, universally recognized visual language for engineering drawings.

Essential electrical wiring symbols.

1 Conductors and connections

Line/wire: A simple solid straight line representing an electrical conductor.

Connection/Terminal: A solid dot indicating where two or more lines are joined.

Terminal Connector: A small circle on a conductor indicating a junction point or test point.

2 Switches and controls

Normally open switch (NO): A simple break in the line, showing an open circuit.

Normally closed switch (NC): A slanted line connecting two dots, indicating a closed circuit.

3 Protection and grounding devices

Earth (ground): A vertical line ending in three horizontal lines that get smaller downward.

Fuse: A curved line or a straight line cutting through a rectangular box.

Task	Questions Description	Marks
3	a) Describe the principles of operation of various control systems	5
	b) Describe the principles of operation of single and three phase circuit breakers and core balance earth leakage relays (wound primaries and straight primaries with tripping relay)	5

(a) Open-loop control systems: These systems operate solely based on input commands without using feedback to check the actual output. The controller operates the actuator according to a predefined schedule.

Closed-loop control systems: Also known as automatic control, this system uses sensors to monitor the output continuously. The system calculates the error (difference between desired and actual output) and uses that information to adjust the actuator, reducing or eliminating the error.

(b) Circuit breakers principle of operation

Single-phase circuit breaker: Uses a bimetallic strip to bend under prolonged heat (overload) or an electromagnet to pull a latch (short circuit), causing a spring-loaded contact to open.

Three-phase circuit breaker: Operates similarly to single-phase but features three independent sets of contacts that are linked internally. A fault in any single phase trips all three contacts simultaneously, protecting against unbalanced loads and phase loss.

Wound primaries: The live and neutral wire are wound as separate coils around the core.

5 + 14

LEARNING UNIT 3:

KM-05-KT03: Earthing and bonding (30%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
4	a) Describe the earthing systems relevant to SANS 10142-1.	3
	b) Describe how a common earth electrode is used in reticulation circuits	3
	c) Describe the provision of earthing for underground cables and overhead lines	4

TN-S system: The safest TN system, with separate neutral (N) and protective earth (PE) conductors running from the transformer to the installation.

TN-C-S system (preferred in SA): Known as multiple earthed neutral (MEN), where the supply is TN-C (combined PEN), but splits into separate N and PE within the installation.

3 TT system: The installation has its own earthing electrode, independent of the supply source. This is used when the utility does not provide an earth connection.

TN-C system (not approved): The neutral and earth are combined throughout. Not permitted for low-voltage installations in SA due to risks if the PEN conductor breaks.

(3) Protection mechanism: It acts as a safety layer for equipment and people by allowing fault currents to dissipate safely, preventing electrical shock and reducing the risk of fire.

System stability: In rectification circuits, it ensures voltage stability and ensures that in the event of a fault (e.g. short circuit), the protection system effectively triggers.

Implementation: It is typically a buried conductor (like copper-bonded rods) used in IT or TN systems, often connected to the main earthing terminal (MET) of a building's electrical setup.

Component of earthing system: It is used for connecting to foundations, as a ring conductor, and with lightning protection systems to guarantee safety across the site.

Influence of environment: The effectiveness of the electrode is heavily dependent on soil resistivity; it often requires deep burial to maintain a low-resistance path, particularly in rocky areas.

(c) Earthing for underground cables

Metallic sheath/Armour earthing: The metallic sheath or armor of the cable is connected to the ground at one or both ends (single or multi-point earthing) to ensure the cable remains at ground potential.

Earth electrodes: Earth rods or plates are driven into the ground to provide a definitive path for fault current dissipation, especially at cable joints and terminals.

Cable termination earthing: At substations, all cable sheaths are bonded to the main substation earth grid.

Earthing overhead lines

Earth wire (Shield wire): A galvanized steel wire is attached at the highest point of the tower/pole structure to intercept lightning strikes.

Tower/pole grounding: The support structure (tower or pole) is connected to a dedicated earth electrode, such as a ground rod driven into earth, at each structure location to transfer lightning currents safely.

Assessor's Report

Candidate's Name	<i>Munyizi Ndabene</i>	ID No.	<i>250115587089</i>
Assessor's Name		Reg. No.	
Title	<i>Wine waysand writing</i>		
ASSESSMENT DECISION			
Outcome	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
	<i>H.T.D.</i>		
Signature of Assessor	Signature of Candidate		

Candidate Feedback Report

Candidate's Name	Muyisi Ndabeni		ID No.	950115587089
Assessor's Name		Reg. No.		
Title	Wire ways and Wiring			
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓		Good	
Overall Assessment Decision				
Additional Notes				
Date				
				
Signature of Assessor			Signature of Candidate	

Candidate Appeal Form

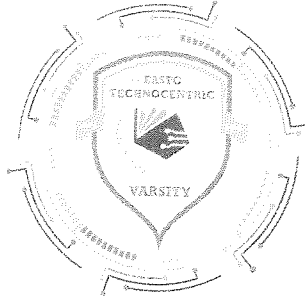
Candidate's Name:		ID No.	
Assessor's Name:		Reg. No.	
Title:			
Date:			
SECTION 1			
Candidate's reason for disagreeing with the assessment decision			
Assessor's rationale for the assessment decision			
Candidate's signature			
Assessor's signature			
SECTION 2			
Moderator's decision			
Moderator's signature			

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Munyisi Ndabeni	ID No.	9501115587089
Title	Wire ways and wiring		
MODERATION DECISION			
Outcome	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>Munyisi</i>		NAME of ASSESSOR	
VENUE <i>EASTC</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
Learner	Date	Assessor	Date
		Moderator	Date



M. Ndabeni

44 / 78 = 95%
30
93%

Wire ways and wiring, FORMATIVE ASSESSMENT

Module #	671101000-KM-05:
NQF Level	level 4,
Notional hours	110
Credit(s)	Cr 11
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	Mvungisi Ndabeni
Contact Address	433 Tswelopele section, Tembisa
Telephone (H)	
Telephone (W)	
Facsimile	
Cellular	0838004724
E-mail	mvudabeni@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓
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Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
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1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources

Barriers

People to be involved with assessment

Learner: <i>Mwajisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Mwajisi Ndabeni*

Learners Name: *Mwajisi Ndabeni*

Learner's signature: *[Signature]*

Date:

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Mungisi Ndabeni</i>	Signature <i>[Signature]</i>
Facilitator Name:	Signature:

Complete the following questions as per instructions provided



LEARNING UNIT 1:

KM-05-KT01: Wire ways (35%)

Learning Outcome 1:	KT0101 Definition, types, purpose and applications of wireways
Learning Outcome 2:	KT0102 General provisions for wireways as prescribed in the SANS 10142-1

Task	Questions Description	Marks
1.	a) Describe and explain the provisions in the SANS 10142-1 Code of Practice for the correct and safe installation and use of the various types of wireways	5
	b) Describe factors that influences the selection of wireways	5

Mechanical protection and material: Wireways must be made of materials that are non-combustible and resistant to corrosion to ensure long-term integrity.

Capacity and overcrowding: The wireway, whether conduit or trunking, must have sufficient space for the cable to prevent overcrowding, which can cause ^{heat} buildup and insulation damage.

Support and fixing: Wireways must be securely fixed to prevent movement. They should be supported at intervals not exceeding 1.5m to maintain structural integrity.

Environmental protection: Wireways must be chosen based on environmental conditions (e.g. moisture, chemicals), ensuring they maintain the necessary IP (Ingress Protection) rating for the area.

Bonding and earthing: All metal wireways (steel conduits, metal trunking, cable trays) must be electrically continuous and bonded to the main earthing system to

ensure safety against electric shock.

Separation of circuits: Different systems (e.g. power and telecommunications) must be adequately separated or shielded to prevent interference, as per SANS 10142-1 guidelines.

Accessibility: Wireways should be installed in locations that are accessible for maintenance and inspection.

(b) Environmental conditions: The surrounding environment dictates the material (e.g. galvanized steel for dry, stainless steel for harsh/corrosive, PVC for wet areas).

Protection levels: Wireways must have appropriate NEMA ratings to protect against dust, water spray, or oil ingress.

Capacity and size: The wireway cross-sectional area must accommodate the total number of cables and allow for proper fill ratios, including heat dissipation for high-load, large-diameter cables.

Installation method: Whether the wireway is installed overhead, underground, or along walls impacts the support structure, layout, and mounting requirements.

Accessibility and maintenance: The type of wireway (e.g. lay-in vs pull-through) depends on if cables need to be easily added, removed or inspected later.

Future expansion: It is crucial to choose a size that permits future additions to the cabling system without requiring a complete redesign.

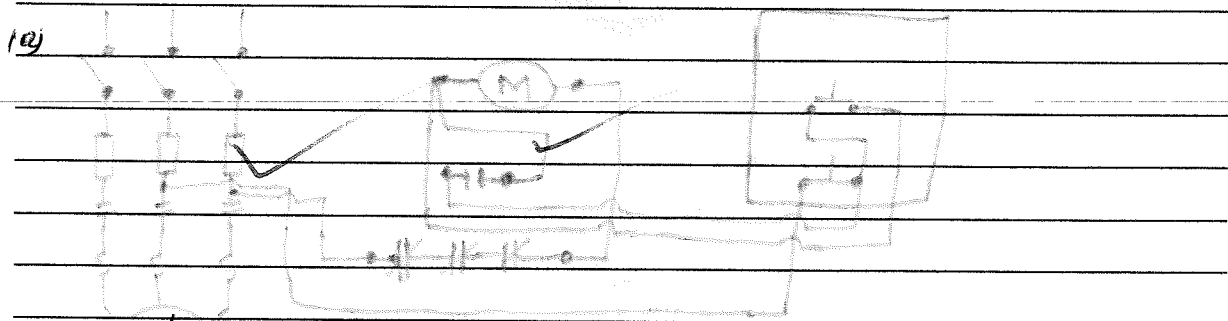
5

LEARNING UNIT 2:

KM-05-KT02: Wiring of installations (35%)

Learning Outcome 1:	KT0201 Regulations and statutory requirements for wiring of premises
Learning Outcome 2:	KT0202 Electrical diagrams and symbols
Learning Outcome 3:	KT0203 Electrical components and their applications

Task	Questions Description	Marks
2	a) Identify electrical components and draw schematic diagrams of installations	3
	b) State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage protection unit	2



Key components include power sources, circuit breakers, switches, and loads (lights/motors) with lines indicating wires.

b) Earthing, fuses, circuit breakers, and earth leakage units (RCDs/ELEBs) provide essential electrical safety by preventing electric shocks, fires and equipment damage. They work together to ground fault currents, breaker currents during overloads and detect minute leakage, automatically isolating power to protect people and property.

Task	Questions Description	Marks
3	Describe the purpose of load distribution, lightning arrestors and energy control units.	5

Lightning arrestors (Surge arrestors): Protect power lines, transformers and electrical equipment from damage caused by lightning strikes or voltage transients. During a high-voltage event, it provides a low-resistance path to the ground, diverting surge energy away from critical component before returning to a high-resistance state.

Energy control units: Manages electrical loads, typically by shedding non-essential loads (like heating/air conditioning) during peak demand periods to prevent system overload. Acts as an automatic load-shedding device that switches off certain devices when energy demand becomes too high, ensuring overall grid or household power stability.

Load distribution (General engineering context): Ensures forces, weights, or electrical loads are distributed evenly across a system (e.g. structural beam or electrical circuit) to maintain stability and prevent failure.

11 (5)

LEARNING UNIT 3:

KM-05-KT03: Earthing and bonding (30%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
4	a) Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation	2
	b) Explain the term 'systems earthing'	3

(a) Regulatory requirements mandate that the neutral conductor is earthed at the source (supplier side) for voltage stability and fault path creation, while consumer-side earthing focuses on safety bonding at the main service panel to ensure protective device operation. Key standards (e.g. NEC, IEC, SANS) prohibit neutral-ground bonding in subpanels to prevent stray currents and demand dedicated earth electrodes for TN-C-S/TT systems.

(b) A systems earthing arrangement (or grounding system) connects the conductive parts of an electrical installation to the earth, providing a low-resistance path for fault currents. It protects people from electric shocks and prevents fire by forcing protective device (like fuses or RCDs) to break the circuit during a fault.

Task	Questions Description	Marks
4	Explain the various processes of measuring, testing and calculating earthing and bonding system values.	5

1. Measuring earth electrode resistance

Fall-of-potential method (3 pole): The standard method involving driving two auxiliary spikes into the soil, ideally 20m to 30m from the earth electrode, and driving current through them to measure resistance.

Stateless measurement: Use two clamps around the ground rod to induce a voltage and measure current, calculating loop resistance in multi-grounded systems without disconnecting the ground.

2. Testing earthing and bonding

Bonding continuity test: Verifies the resistance between main/supplementary equipotential bonding and the main earth terminal is low.

Insulation resistance testing: Ensures that live conductors are not taking current to the earthing system. A test voltage (500V or 1000VDC) is applied.

3. Calculating earthing and bonding values

Earth fault loop impedance equation: $Z_s \leq \frac{U_0}{I_a}$

Where U_0 is the nominal voltage to earth and I_a is the current causing automatic disconnection.

Electrode resistance formula: for a single electrode, $R = \rho \div (2\pi L)$, where ρ is soil resistivity and L is length.

Bonding conductor sizing: Calculated base on the maximum potential fault current and allowable temperature rise of the conductor.

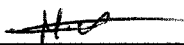
5

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

Outcomes		1	2	3
1	KT0201 Regulations and statutory requirements for wiring of premises			✓
2	KT0202 Electrical diagrams and symbols			✓
3	KT0303 Regulatory and statutory requirements related to earthing and bonding			✓



Learner Signature

Date

Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT

Learner's Total Mark:

Requirements met ☐

Requirements not met ☐

Action/s required:

By when:

LEARNER FEEDBACK & COMMENTS

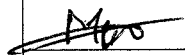
Good work

DECLARATION BY THE FACILITATOR

I (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER

I, Munyisi Ndabw.....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.

**Learner****Date****Facilitator****Date****DECLARATION BY THE ASSESSOR**

I (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.

Assessor**Date****Moderator****Date**

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire
2. Observation : As per Practical Assessment Guide
3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

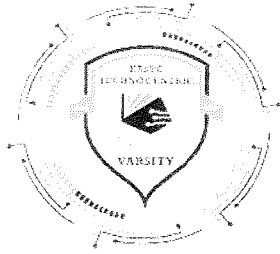
What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments

4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit



LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 06	Rotating Electrical Machinery, NQF Level 4.	13

M. Ndabeni

58
69
84%

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ROTATING ELECTRICAL MACHINERY FORMATIVE ASSESSMENT

Details	Please Complete this Section
Name	Munyisi
Surname	Aldabeni
Organisation:	EASTC TECHNOCENTRIC VARSITY
Unit/Dept:	ELECTRICAL ENGINEERING
Facilitator Name:	

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Ndabeni*

Learners Name: *Munyisi Ndabeni*Learner's signature: *Am* Date: _____

Facilitator Name: _____

Facilitator's signature: _____ Date: _____

Complete the following questions as per instructions provided

LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction	5
	b) Describe with the aid of diagrams, the configuration of motor connections	5
	c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	10

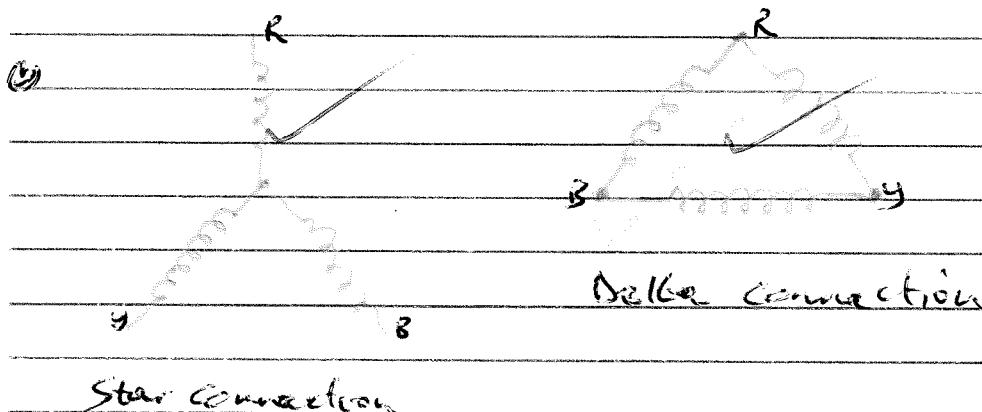
Squirrel cage induction motor: Consists of a stator with laminated iron cores holding three-phase windings. The rotor is a cylinder with conducting bars arranged in slots, short-circuited at both ends by end rings resembling a cage. Wound rotor induction motor: The stator is similar to a cage motor but the rotor has a three-phase winding similar to the stator. The stator windings are connected to slip rings on the shaft, allowing external resistance to be added. (5)

Synchronous motor: The stator carries a three-phase winding. The rotor requires an independent excitation system, such as permanent magnets or a D.C.-excited field winding to create a constant magnetic field.

(7)

(j) Motor connections define how motor windings are configured to a power source, primarily in star (Y) or Delta (Δ) arrangements for 3-phase AC motors or series/parallel for DC/dual-voltage motors.

(5) (i) Overload protection devices use a combination of current rating (tripping threshold) and time delay (inverse-time characteristic) to allow safe motor startup while providing fast protection against sustained overloads and high-intensity short circuits. Time delay prevents nuisance tripping from high inrush currents, while specific current ratings detect locked rotors or operational faults, preventing winding damage. The current rating determines when the device starts to react and the time delay determines how long it waits to trip based on the intensity of the overcurrent.



LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of alternating current motors	4
	b) Describe and explain how tests are conducted on alternating current motors	4
	c) Describe and explain protection devices used for alternating current motors	5
	d) Compare the advantages and disadvantages of single- and three-phase motors	6

AC motors operate on the principle of electromagnetic induction, where alternating current (AC) applied to the stationary stator create a rotating magnetic field (RMF), cuts across the rotor conductors, inducing a current that generates its own mag.

3
magnetic field, causing the rotor to spin in the same direction to catch up with the R.M.F.

10
(b) A.C motor testing involves evaluating electrical and mechanical integrity via offline static tests (resistance, insulation, surge) and online dynamic tests (vibration, power quality). Key procedures include checking winding continuity/balance, measuring insulation resistance to ground, and testing under load to ensure performance without failure.

3
(c) A.C motor protection devices prevent ~~doing~~ damage from electrical faults or abnormal conditions, ensuring longevity and safety. Key devices include overload relays, fuses/circuit breakers and thermal protectors, which monitor current, temperature, and voltage to disconnect power during overcurrent, short-circuits, phase failure, or overheating.

(d) Single-phase motors: Advantages: lower initial purchase price, operates on standard residential single-phase power, simple construction, ideal for low-power, small-scale applications.
Disadvantages: Lower efficiency and lower starting torque, not self-starting, requiring capacitors or auxiliary winding, higher vibration and noise, generally limited to smaller loads.
Three-phase motor: Advantages: Higher efficiency, superior starting torque, self-starting with no additional, complex starting components, higher power density, more consistent smooth delivery.
Disadvantage: Higher initial cost, requires three-phase power supply, which may not be available in residential areas, more complex, making repairs potentially more involved.

LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201	Construction of direct current motors
Learning Outcome 2:	KT0202	Principles of operation for direct current motors
Learning Outcome 3:	KT0203	Configuration of direct current motor connections
Learning Outcome 4:	KT0204	Types and application of direct current motors
Learning Outcome 5:	KT0205	Testing principles of direct current motors
Learning Outcome 6:	KT0206	Protection of direct current motors
Learning Outcome 7:	KT0207	Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) List the different types of direct current motors and describe their construction	2
	b) Describe the applications of different types of direct current motors	2
	c) Describe with the aid of diagrams, the configuration of motor connections, power circuits and control circuits.	2
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation, short circuits.	2
	e) Calculate direct current motor properties	2

(a) Permanent magnet DC motors: Uses permanent magnets (often rare earth materials) on the stator to create the magnetic field. The rotor is an armature wound with copper coils connected to a commutator.

Series DC motors: The field winding (stator) is connected in series with the armature winding (rotor). The field

(2)

coil is made of a few turns of thick wire to handle high armature current.

Shunt DC motors: The field winding is connected in parallel (shunt) with the armature. The field has many turns of thin wire to maintain high resistance and low field current.

Compound DC motors: Combines both shunt and series field windings on the stator. It can be long-shunt or short-shunt.

(b) DC series motor (high starting torque, variable speed): Used in electric locomotives (traction systems), cranes, hoists, winding systems, conveyors and heavy-duty electric vehicles.

DC shunt motor (constant speed, low starting torque): Used in centrifugal pumps, fans, blowers, centrifugal pumps, conveyors, lathes, spinning machines, and paper machines.

(2) DC compound motor (high torque and relatively constant speed): Used in presses, shears, elevators, rolling mills, heavy planners and grinders.

Permanent magnet DC motor: found in portable tools, electric toothbrushes, computer drives, robotics, food mixers, car wipers and computer drives.

(c)

(d) Time Delay (inverse time characteristic)

~~Start delay~~

Startup/transient overloads: Time-delay (inverse-time) devices allow high startup current for several or several seconds without tripping, preventing nuisance trips.

Operational overloads: Time and a moderate delay is needed to handle temporary surges.

Locked rotor: Time delay must be short enough to prevent heat from exceeding the motor's tolerance, usually achieved by a fast-acting, heat-activated mechanism.

Current rating (set point)

Overload protection: Should be matched closely to the motor's full load current (FLC), often with a service factor, to detect small increases in current that, overtime, would destroy insulation.

Locked Rotor/short circuit: Must be high enough to allow 600-900% inrush current during start-up, but low enough to trip before the high current "locked-rotor" condition damages windings.

(e) for 24 V motor with 0.33 Ω resistance and 38.5 A armature current: $P_{input} = 8360 \text{ W}$ and total loss is 1037 W

$$\text{EMF: } E_b = 24 \text{ V} - IR = 24 - (38.5 \times 0.33) = 11.28 \text{ V}$$

$$P = I_a R_a = (38.5)^2 \times 0.33 = 489 \text{ W}$$

$$P_{out} = \pi \text{ W: } P_{out} = 8360 - 1037 = 7323 \text{ W}$$

$$\text{Efficiency} = \frac{7323}{8360} = 87.6\%$$

LEARNING UNIT 3:

Rotating electrical machinery - Alternators and Generators (30%)

Learning Outcome 1:	KT0301 Construction of alternators and generators
Learning Outcome 2:	KT0302 Principle of operation of alternators and generators
Learning Outcome 3:	KT0303 Configuration of alternator and generator connections
Learning Outcome 4:	KT0304 Types and application of alternators and generators
Learning Outcome 5:	KT0305 Testing principles of alternators and generators
Learning Outcome 6:	KT0306 Protection of alternators and generators
Learning Outcome 7:	KT0307 Calculate properties of alternators and generators

Task	Questions Description	Marks
	a) Explain the principle of operation of alternators and generators	6
	b) Describe the applications of different types of alternators and generators	4
3	c) Describe and explain protection devices used for alternators and generators	4
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting alternators and generators from damage in the case of locked rotors, overload during operation and short circuits.	6

*) Alternators and generators operate on Faraday's law of electromagnetic induction, converting mechanical energy into electrical energy by rotating a magnetic field within wire coils (stator) or vice versa. The relative motion induces a voltage, producing alternating current (AC).

Alternator principle: Alternators feature a rotating field (rotor) inside a stationary armature (stator). As the rotor spins, it creates a magnetic field that cuts across the stator windings, generating 3-phase AC, which is then converted into DC by a rectifier and regulated by a voltage regulator.

Generator principle: Traditional generators (DC) operate similarly but often use a commutator to produce direct current, whereas alternators produce AC.

(b) Automotive alternators: Used in cars, trucks and vehicles to charge the battery and power electrical systems (headlights, radio, AC) while running.

Brushless alternators: Standard in modern diesel and gas generators, offering high reliability for backup power systems.

Marine alternators: Specially designed to resist saltwater and corrosion in boats and ships.

Locomotive alternators: Used in diesel-electric locomotives to power traction motors.

(c) Differential protection: This is the primary protection for stator winding faults. It works by comparing current transformers (CTs) at both ends of each phase windings. If the input current does not equal the output current, a fault exists within the zone, and the relay trips the circuit breaker.

Overcurrent and circuit protection: These devices, such as circuit breakers and relays, detect when the current exceeds set limits (due to overload or short circuit) and interrupt the circuit to protect the windings and insulation from burning.

Stator earth fault protection: Since stator neutrals are often grounded through resistors, these relays detect leakage current to the ground, protecting against

stator winding insulation failure.

Rotor earth fault protection: These detect insulation failure in the rotor field circuit using methods like DC or AC injection, as a single earth fault may not trip the machine but a second one could cause severe imbalances.

(d) Overload during operation: The device's current rating is set slightly above the nominal operating current (e.g. 120% - 25% higher). The time delay is essential to allow for normal, safe, and temporary overloads (like starting heavy loads) without interrupting service, while acting before the insulation exceeds safe operating temperatures.

Locked Rotor (stalled) conditions: Since a locked rotor draws very high current (often 6x the rated current), the protection device must have a short, precise time delay, often provided by instantaneous settings, to cut power before the generator or motor suffers significant heat damage.

Short circuits: Under short-circuit conditions, extreme current demands an instantaneous, almost zero-time delay response (e.g. via a circuit breaker or fast-acting fuse) to avoid massive magnetic forces and heat damage to windings.

Influence on protection: A low current rating and long time delay would lead to nuisance trips. Conversely, a high current rating with a long time delay risks damaging the equipment before the protection activates.

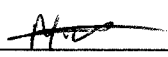
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SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	KT0301 Construction of alternators and generators			✓
2	KT0302 Principle of operation of alternators and generators			✓
3	KT0303 Configuration of alternator and generator connections			✓
4	KT0304 Types and application of alternators and generators			✓
5	KT0305 Testing principles of alternators and generators			✓
6	KT0306 Protection of alternators and generators			✓
7	KT0307 Calculate properties of alternators and generators			✓



Learner Signature

Date



Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

Good work

ASSESSMENT JUDGEMENT

Learner's Total Mark:

Requirements met ☒

Requirements not met ☐

Action/s required:

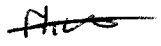
By when:

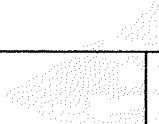
LEARNER FEEDBACK & COMMENTS

Good work

DECLARATION BY THE FACILITATOR

I Balennah Tyabadi (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, <u>M. Nuyisi Ndabeni</u>declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
			
Learner	Date	Facilitator	Date

DECLARATION BY THE ASSESSOR			
I, <u>Baenmah Tyabadi</u> (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
			
Assessor	Date	Moderator	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire
2. Observation : As per Practical Assessment Guide
3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments
4. Conduct observation in workplace where applicable

5. Complete learner toolkit

6. Complete assessor toolkit



M. Ndabeni

33
40
83%

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ROTATING ELECTRICAL MACHINERY SUMMATIVE ASSESMENT

Details	Please Complete this Section
Name & Surname:	Munyira Ndabeni
Organisation:	EASTC TECHNOCENTRIC VARSITY
Unit/Dept:	ELECTRICAL ENGINEERING
Facilitator Name:	

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Ndabeni*

Learners Name: *Munyisi Ndabeni*Learner's signature: *[Signature]* Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Munyisi Ndabari</i>	Signature <i>HTO</i>
Assessors Name:	Signature:



INSTRUCTIONS:

The following sections will test your knowledge, and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

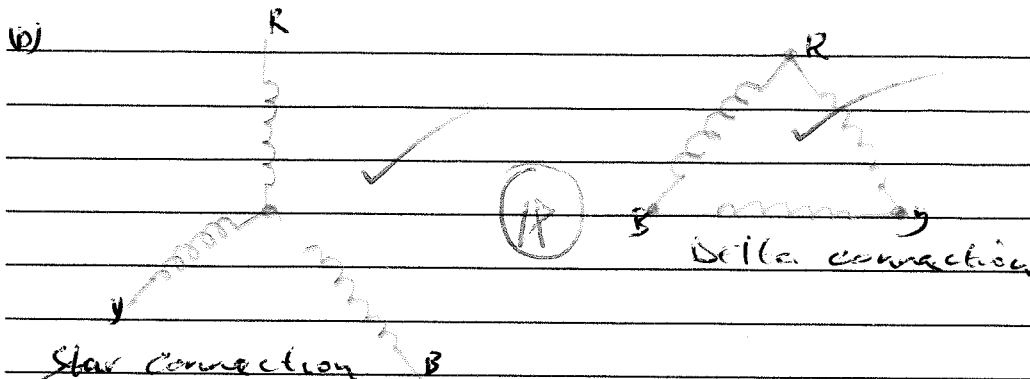
Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction	5
	b) Describe with the aid of diagrams, the configuration of motor connections	5
	c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	10

(a) Squirrel cage induction motor: Consists of a stator with laminated iron cores holding three-phase windings. The rotor is a cylinder with conducting bars arranged in slots, short-circuited at both ends by end rings resembling a cage.

Wound rotor induction motor: The stator is similar to a cage motor, but the rotor has a three-phase winding similar to the stator. The rotor windings are connected to slip rings on the shaft, allowing external resistance to be added.

Synchronous motor: The stator carries a three-phase winding. The rotor requires an independent excitation system, such as permanent magnets or a DC-excited field winding to create a constant magnetic field.



Motor connections define how motor windings are configured to a power source, primarily in star (V) or Delta (W) arrangements for 3 phase AC motors or series/parallel for DC/dual-voltage motors.

a) Overload protection devices use a combination of current rating (tripping threshold) and time delay (inverse-time characteristic) to allow safe motor startup while providing fast protection against sustained overload and high-intensity short circuits. Time delay prevents nuisance tripping from high inrush currents, while specific current ratings detect locked rotors or operational faults, preventing winding damage.

The current rating determines when the device starts to react and the time delay determines how long it waits to trip based on the intensity of the overcurrent.

LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201 Construction of direct current motors
Learning Outcome 2:	KT0202 Principles of operation for direct current motors
Learning Outcome 3:	KT0203 Configuration of direct current motor connections
Learning Outcome 4:	KT0204 Types and application of direct current motors
Learning Outcome 5:	KT0205 Testing principles of direct current motors
Learning Outcome 6:	KT0206 Protection of direct current motors
Learning Outcome 7:	KT0207 Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of direct current motors	2
	b) Describe and explain how various tests are conducted on direct current motors	2
	c) Describe and explain protection devices used for direct current motors	3
	d) Compare the advantages and disadvantages of different types of direct current motors	3

② A DC motor operates on the principle that a current-carrying conductor, when placed within a magnetic field, experiences a mechanical force (Lorentz force). This force determined by Fleming's left-Hand Rule ($F \propto BIL$), causes the armature (rotating part) to turn, converting electrical energy into mechanical rotation.

④ DC motor testing methods include basic multimeter checks for continuity and winding resistance, insulation tests (Megger) to detect short circuits, and functional tests like no load or load tests.

Key procedures include measuring armature resistance, checking brush commutator connections, and using the Swinburne or Hopkinson test for efficiency.

15 Thermal or load relays: Protect against continuous overloading, which leads to overheating. They use bimetallic strips that bend when hot, tripping the motor when the current exceeds set limits for too long.

3 Circuit breaker and fuses: Provide instantaneous protection against severe overcurrents, such as short circuits or earth faults, by rapidly breaking the circuit to prevent fire or damage.

Loss-of-field relays: Specifically critical for DC shunt or compound motors. If the current is lost, the motor can experience dangerous overspeeding. A relay monitors the field circuit and shuts down the motor if the field current drops below a safe threshold.

16 Brushed DC motors

Advantages: Low cost, simple control, high starting torque.

Disadvantages: Brushes wear out, requiring maintenance; low efficiency; produce electrical noise.

Brushless DC motors

Advantages: High efficiency, long lifespan, low maintenance, quiet operation, and high power density.

Disadvantages: Higher initial cost, requires complex electronic controllers.

3

LEARNING UNIT 3:

Rotating electrical machinery - Alternators and Generators (30%)

Learning Outcome 1:	KT0301 Construction of alternators and generators
Learning Outcome 2:	KT0302 Principle of operation of alternators and generators
Learning Outcome 3:	KT0303 Configuration of alternator and generator connections
Learning Outcome 4:	KT0304 Types and application of alternators and generators
Learning Outcome 5:	KT0305 Testing principles of alternators and generators
Learning Outcome 6:	KT0306 Protection of alternators and generators
Learning Outcome 7:	KT0307 Calculate properties of alternators and generators

Task	Questions Description	Marks
2	a) List the different types of alternators and generators and describe their construction	2
	b) Explain the fundamental differences between alternators and generators	2
	c) Describe and explain how tests are conducted on alternators and generators	3
	d) Describe with the aid of diagrams, the configuration of alternator and generator connections, power circuits and control circuits.	3

Salient pole generator/alternator : features large, projecting poles used for low-speed operation.

Cylindrical Rotor Alternator (Tubo alternator) : features a smooth, cylindrical rotor used for high-speed operation.

Brushless alternator : uses a rotating rectifier to eliminate brushes for higher reliability, commonly in automotive and backup power.

Construction

Stator (Armature Core): An iron ring laminated with silicon steel (to minimize eddy currents) containing slots. Armature windings are placed in these slots to generate current.

Cylindrical rotor (field system): A solid steel cylinder with slots for winding, designed for high mechanical strength at high speeds.

Exciter (for Alternators): A small DC generator or rectifier mounted on the shaft to supply DC power to the rotor field windings.

② An alternator is a device that converts mechanical energy into AC electrical energy. A generator is a mechanical device which converts mechanical energy to either AC or DC electrical energy.

① Alternator and generator testing ensures the electrical system produces correct voltage using multimeter or oscilloscope.

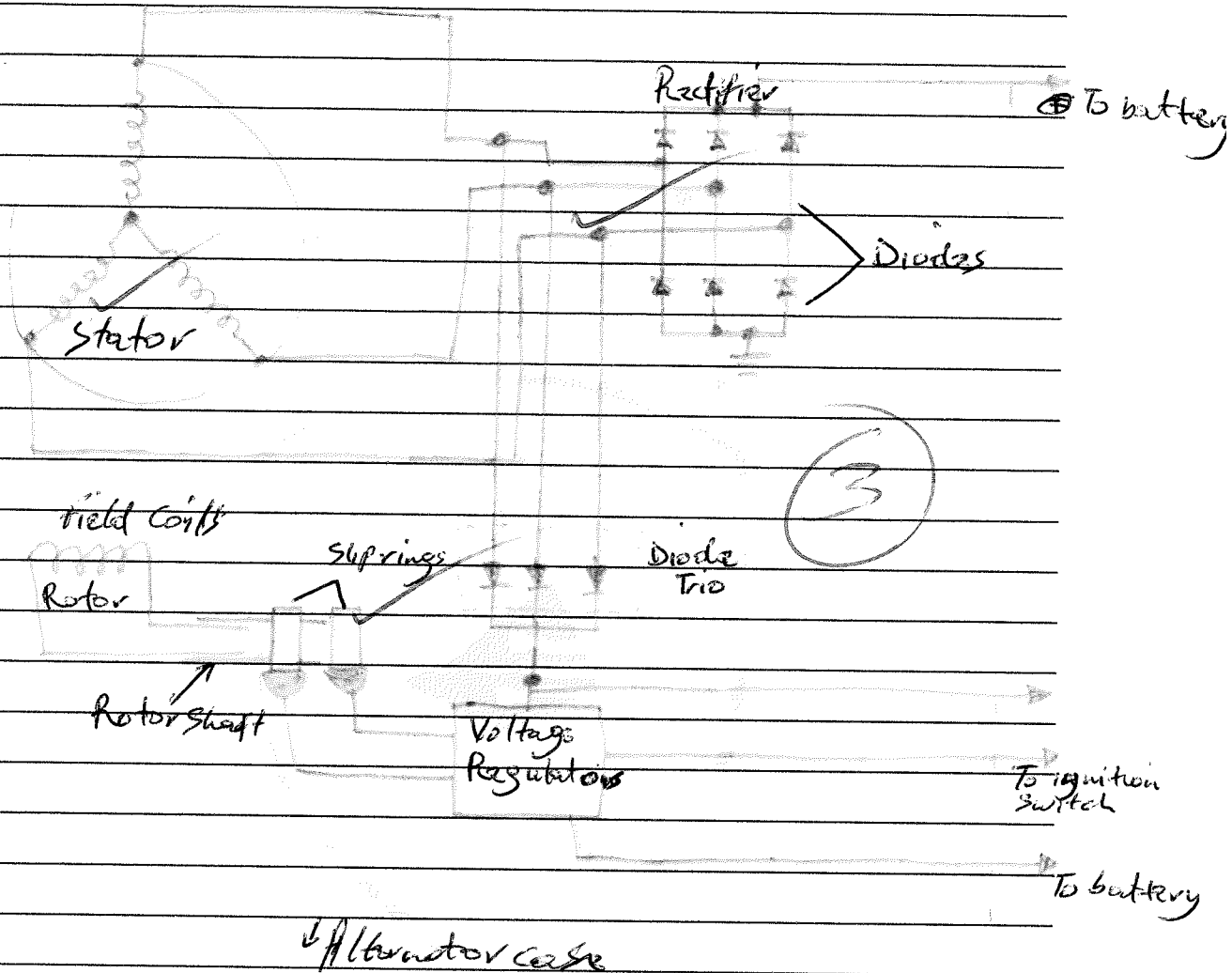
Key test Key testing methods for alternator/generators.

Multimeter Voltage Test: The most common method, done by setting a multimeter to DC voltage. A healthy alternator should show 13.8 to 14.6 volts with the engine running, whereas a fully charged battery alone sits around 12.6 V.

③ **Load test:** This tests the alternator's capacity under stress. The engine running, turn on high-draw accessories like headlights, radio, and air conditioning. Voltage should stay above roughly 13.6 volts, if it drops below, the alternator is struggling.

AC Ripple Test: An alternator should produce DC, but diodes can fail. Setting a multimeter to AC voltage can detect "ripple" (leakage). Higher ripple indicates failed diodes or stator windings.

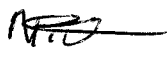
10) Alternator



Assessor's Report

Candidate's Name	<i>Munyira Ndabeni</i>	ID No.	950115587089
Assessor's Name		Reg. No.	
ASSESSMENT DECISION			
	C	NYC	Comments
KT0101 Construction of alternating current motors	✓		Good
KT0302 Principle of operation of alternators and generators	✓		Good
KT0303 Configuration of alternator and generator connections	✓		Good
KT0304 Types and application of alternators and generators	✓		Good
KT0305 Testing principles of alternators and generators	✓		Good
KT0306 Protection of alternators and generators	✓		Good
KT0307 Calculate properties of alternators and generators	✓		Good
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
	<i>[Signature]</i>		
Signature of Assessor	Signature of Candidate		

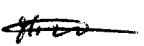
Candidate Feedback Report

Candidate's Name	Munyisi Ndabeni	ID No.	950115587089
Assessor's Name		Reg. No.	
ASSESSMENT DECISION			
Source of Evidence	C	NYC	Comment
Knowledge questions	✓		Good
Product /Work sample	✓		Good
Project	✓		Good
Indirect Evidence	✓		Good
Overall Assessment Decision			
Additional Notes			
Date			
			
Signature of Assessor		Signature of Candidate	

Candidate Appeal Form

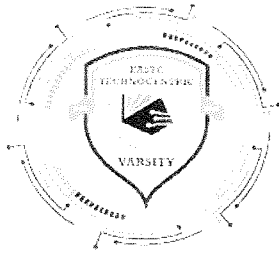
Candidate's Name: <i>Munyisi Ndabeni</i>	ID No.	9501115587089
Assessor's Name:	Reg. No.	
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Mvuyisi Ndabeni	ID No.	9501115587089
MODERATION DECISION			
	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>Munisi</i>		NAME of ASSESSOR	
VENUE <i>EASTC</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
Learner	Date	Assessor	Date
Moderator		Date	



LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 07	Electrical Supply Systems and Components, NQF Level 4.	31

M. Ndabeni

16 / 60 = 90

Module #	671101000-KM-07:
NQF Level	level 4,
Notional hours	310
Credit(s)	Cr 31
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

FORMATIVE ASSESSMENT

Details	Please Complete this Section
Name	Ndabeni Mvuyisi
Surname	Ndabeni
Organisation:	
Unit/Dept:	
Facilitator Name:	

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓
Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed	
Training	Classroom training registers	Training Provider Date:		
Self assessment	Assessment contract signed & dated	Learner file Date:		
Assessment contract	Assessment contract signed & dated	Learner file Date:		
Initial meeting	Assessor briefing checklist	Learner file Date:		
	Assessment contract & assessment plan	Learner file Date:		
Formative Assessment	Assessment instruments	Learner file Date:		
Summative Assessment	Assessment instruments	Learner file Date:		
Other Evidence	Research portfolio (if applicable)	Learner file Date:		
Feedback	Feedback Report	Learner file Date:		
Moderation	Moderators report	Learner file Date:		
Judgement	Assessor Summary Report / Moderator report	Learner file Date:		
1 st Reassessment	Assessors summary report / instruments	Learner file Date:		
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:		

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Adabeni</i>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Adabeni*

Learners Name: *Munyisi Adabeni*Learner's signature: *[Signature]*

Date: _____

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ENT APPEALS PROCEDURE

ider the following circumstances
t procedures were not followed during assessments
e was taken into account during the assessment
ject matter expert or did not have a subject matter
ent process
according to the performance criteria and range
e unit standard
re available for assessment
against a decision of the assessment will lodge such an
the internal moderator within 2 days of the assessment

should complete the "Learner's Notice of Assessment
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re-affirm the assessor's decision, the learner may appeal
days after the initial moderator's feedback session. The
be final. Should the external verifier re-affirm the
or re-evaluation will be upon the learner. Should the
ne assessor's decision, the cost for re-evaluation will be

er instructions provided

in

T APPEAL APPLICATION FORM

MENT APPEAL

erator is hereby requested to discuss the outcome

<u>LEARNING UNIT 1:</u> KM-07-KT01: Concepts, theories and principles of supply Systems (20%)

Learning Outcome 1:	KT0101 Theories and concepts of alternating current
Learning Outcome 2:	KT0102 Fundamental principles of alternating current
Learning Outcome 3:	KT0103 Alternating current generation (distribution systems theory)
Learning Outcome 4:	KT0104 Characteristics and calculations from alternating current waveforms
Learning Outcome 5:	KT0105 Theories, concepts and principles of direct current sources
Learning Outcome 6:	KT0106 Sources of direct current
Learning Outcome 7:	KT0107 Calculations of direct and alternating current circuits
Learning Outcome 8:	KT0108 Construction and operating principles of direct current generators
Learning Outcome 9:	KT0109 Characteristics, sources and generation of renewable energy
Learning Outcome 10:	KT0110 Relevant legislative requirements

Task	Questions Description	Marks
1.	a) Explain the generation and differences between single and three phase alternating current by using wave forms and vector diagrams.	4
	b) Explain by drawing wave form diagrams, the differences between line and phase values.	4
	c) Explain the generation principles of direct current.	4
	d) Calculate power and energy in direct and alternating current circuits	4
	e) Explain with the aid of drawings the operating principles of direct current generators.	4

m

a) Single-phase: Generated by a single coil rotating in a magnetic field (or a single coil stator and a rotating magnet). This produces a single, sinusoidal voltage waveform that alternates between maximum positive and negative values.

~~Three-phase~~: Generated by a single coil rotating in a magnetic field (or a single coil stator and a rotating

Three-phase: Generated by three separate coils (or sets of windings) placed 120° apart on the stator. As the magnet rotates, it induces voltage in each coil, resulting in three separate alternating waveforms that are spaced 120° apart in time or electrically separated.

Waveform (Time-domain)

Single-phase: A single sine wave, often alternating at 60 Hz (in the US) or 50 Hz (EU/Other), representing a single push/pull force.

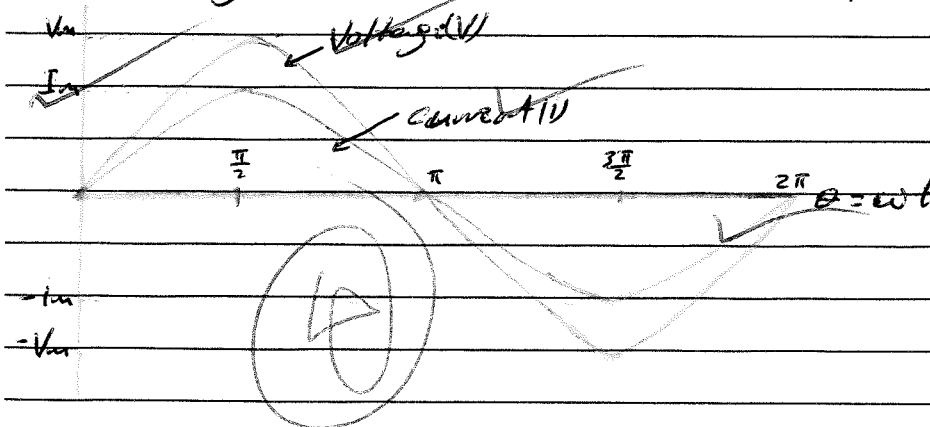
Three-phase: Three distinct sine waves (labeled R, B, C or U, V, W) drawn on the same graph, each following the other by 120° .

Vector diagram (Phasor diagram)

Single-phase: A single vector representing the phase voltage (V) relative to neutral.

~~Three-phase~~: A single vector representing the phase voltage

Three-phase: Three vectors (V_R, V_B, V_C) of equal length, but separated by 120° from one another, commonly shown in a star/wye formation, representing the phase shifts.



Primary cells (Non-rechargeable)

Advantages

Long shelf life: Can be stored for years and used immediately.
Convenience: No charging infrastructure required, instant power.
Lower cost: Cheaper to purchase initially, suitable for low-drain devices.

Safety: Generally lower safety risks regarding overheating compared to rechargeables.

Disadvantages

Not rechargeable: Disposable, leading to high long term cost and waste.

Low load capability: Poor performance for high-power devices.

Environmental impact: Requires careful disposal, not eco-friendly.

Best Applications: Remote controls, toys, wall clocks, flash lights, smoke detectors.

Secondary cells (Rechargeable)

Advantages

Reusable: High cost-effectiveness overtime due to multiple charge cycles.

High power output: Excellent for heavy-duty, high-drain applications.

Eco-friendly: Lower environmental footprint than constantly disposing of Batteries.

Disadvantages

High initial cost: Expensive to purchase, requiring chargers.

Self-discharge: Tend to lose charge overtime, even when not in use.

Limited lifespan: Capacity degrades over many cycles, eventually requiring replacement.

Best Applications: Smartphones, laptops, electric vehicles, power tools, emergency backup systems.

e) Digital multimeter (DMM)

Set the DMM to DC voltage (with a straight line). Touch

the red lead to the positive terminal and the black lead to the negative terminal.

Purpose: Measure open circuit voltage (resting voltage) to quickly determine if a battery is charged or discharged.

Battery load tester

Connect the tester clamps to the battery terminals.

Activate the switch to apply a temporary high load for a few seconds.

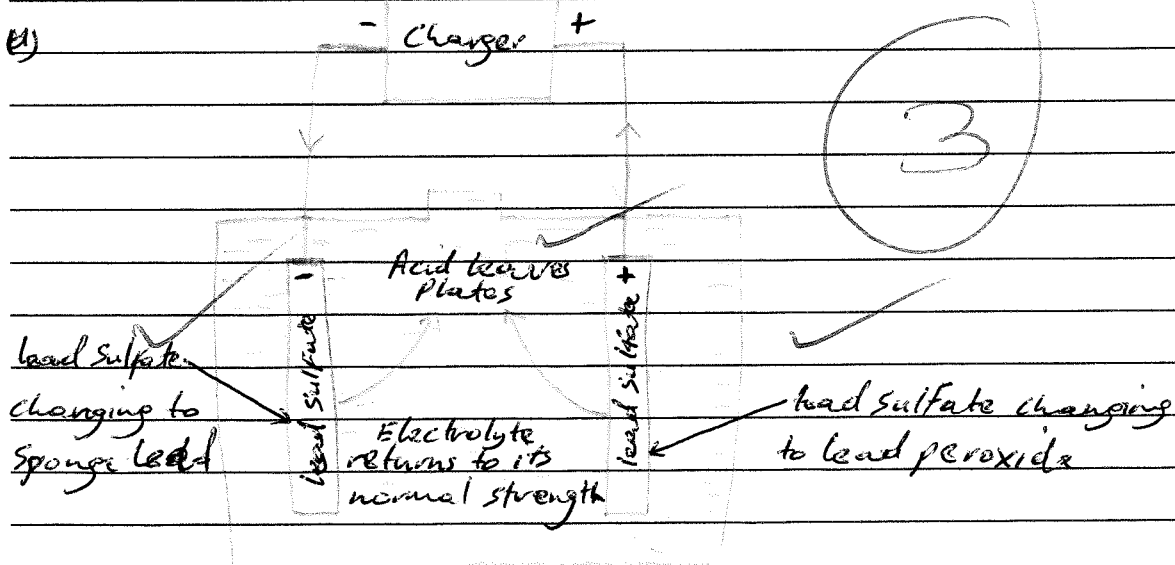
Purpose: Tests the battery's capacity to maintain voltage under heavy load (cranking test), identifying weak or failing batteries that may hold voltage when idle.

Battery analyzer / Analyzer system

Connected to a computer, the analyzer runs pre-set charge/discharge cycles on the battery.

Purpose: Measure true battery capacity, charge acceptance and internal resistance to determine overall battery health and life expectancy.

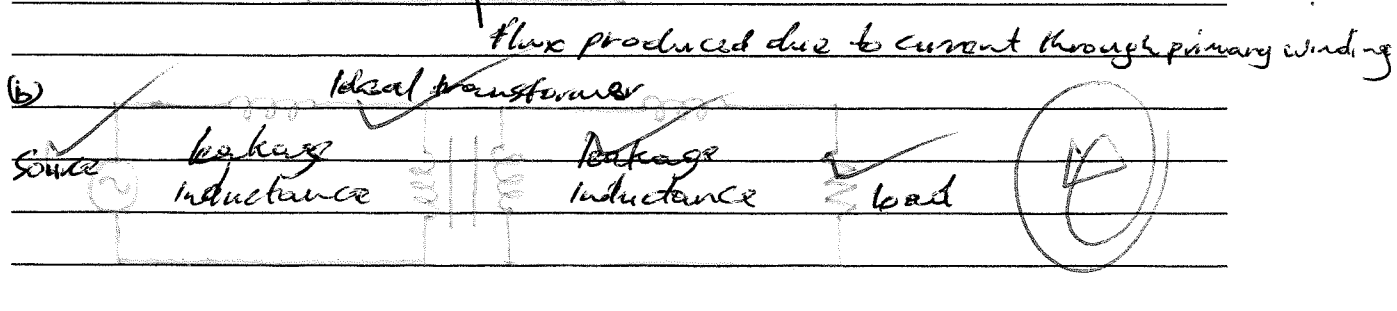
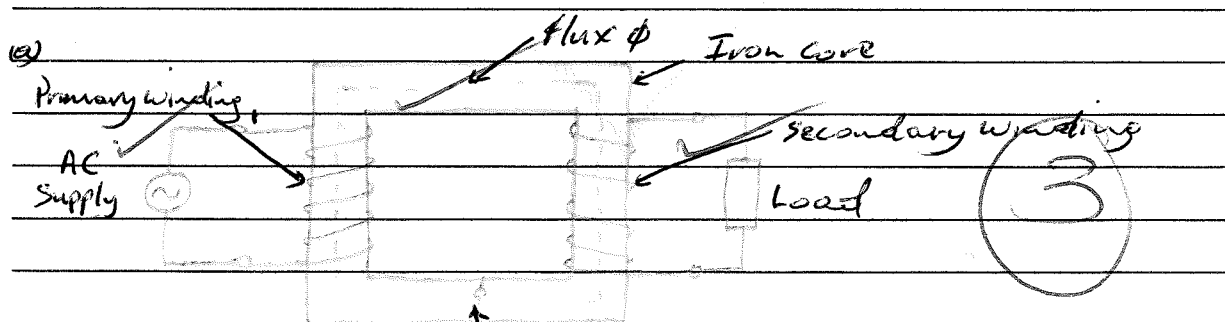
(1)



LEARNING UNIT 3:**KM-07-KT03: Transformers (20%)**

Learning Outcome 1:	KT0301 Theories of single phase wound transformers
Learning Outcome 2:	KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.
Learning Outcome 3:	KT0303 Fundamentals of transformer construction
Learning Outcome 4:	KT0704 Transformer cooling systems
Learning Outcome 5:	KT0305 Principles of single phase transformer operation
Learning Outcome 6:	KT0306 Principles of single phase auto-transformer operation
Learning Outcome 7:	KT0307 Transformer losses
Learning Outcome 8:	KT0308 Formulas and calculations on input and output of transformers
Learning Outcome 9:	KT0709 Maintenance requirements of transformers
Learning Outcome 10:	KT0310 Types of three phase transformers and their applications
Learning Outcome 11:	KT0311 Construction of three phase transformers including open core, closed core and shell or divided core
Learning Outcome 12:	KT0312 Principles of 3 phase transformer operation and configuration
Learning Outcome 13:	KT0313 Principles of 3 phase auto transformer operation and configuration
Learning Outcome 14:	KT0314 Transformer protection and phasing

Task	Questions Description	Marks
3	a) Describe, with the aid of drawings, the construction of single phase transformers	3
	b) Describe, with the aid of drawings, the principle of operation of single phase transformers in terms of mutual inductance and the henry as the unit of inductance	4
	c) Describe types of transformer losses and explain their causes and effects	2
	d) Calculate the terminal voltage, turns and current ratios for an single phase transformer.	3
	e) Describe the various types of construction of three phase transformers - open core, closed core, and shell or divided core	4
	f) Describe the basic transformer maintenance procedures	3
	g) Describe and explain how inductance causes a phase angle between voltage and current and its relationship with the concept 'power factor'	3
	h) Explain the concept of, and need for, transformer tapings	3



(1) Hysteresis loss:

Causes: Occurs as the magnetic domains within the ferromagnetic core align and realign with the alternating magnetic field, creating internal friction.

Effects: Converted into heat energy in the core, reducing efficiency. Reduced by using high-permeability materials like silicon steel.

(2) Eddy current loss

Causes: The alternating magnetic flux induces small, circulating eddy currents within the conductive metal core material.

Effects: Generates heat inside the core, wasting energy. Minimized by laminating the core.

(a) A 2400/240V transformer has a turns ratio of 10.

Voltage ratio: $\frac{2400}{240} = 10$

Secondary voltage (V_2): $V_2 = \frac{N_2}{N_1} V_1 \rightarrow V_2 = \frac{2400}{10} = 240V$

Current ratio ($\frac{I_1}{I_2}$): $\frac{1}{a} = \frac{1}{10}$

Primary current: $I_2 = 100A$ $I_1 = \frac{100}{10} = 10A$

(c) Closed core type: The core consists of three vertical limbs and two yokes forming a continuous rectangular magnetic path.

Shell type / divided core type: The core surrounds the windings, often using "sandwich" or "pancake" windings, where primary and secondary coils are interleaved.

Open core / Core-shell hybrid: While true "open core" (without a return path) is rarely used for 3-phase, 4-limb or 5-limb cores are used to provide a path for leakage flux or to reduce the overall height.

(4) Basic transformer maintenance involves regular, planned inspections and testing to ensure reliability and longevity, typically focusing on visual

(2)

inspections, oil sampling/testing (for oil-filled), insulation resistance testing, and cleaning. Key actions include check for leaks, verifying oil levels, measuring bushing cleanliness, tightening electrical connections, and renewing cooling systems to prevent overheating.

(3) Inductance causes a phase angle in alternating current (AC) circuits because it opposes changes in current, acting as a form of "inertia" that delays the rise and fall of current behind the applied voltage. In an inductive circuit, the current lags the voltage by a phase angle of up to 90 degrees, resulting in a low or "lagging" power factor.

(4) Transformer tapings: Taps are specific access points connected to the windings, usually on the high-voltage (HV) side, allowing a subset of the winding to be used.

Need for transformer tapings:

Voltage regulation: Compensates for voltage drops along transmission lines, maintaining consistent secondary voltage.

(5) Grid fluctuations: Corrects the transformer's output when the primary supply voltage is too high or too low.

Load balancing: Adjust to high peaks loads (evening) or low loads (morning) to keep voltage within the $\pm 10\%$ nominal range.

~~System protection: Adjust to high peak loads (evening) or low loads (morning) to keep voltage~~

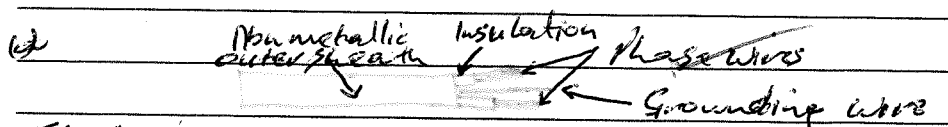
System protection: Ensures electrical appliances operate within their design voltage, preventing failures caused by voltage sags or surges.

LEARNING UNIT 4:

KM-07-KT04: Types of cables and applications (20%)

Learning Outcome 1:	KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.
Learning Outcome 2:	KT0402 Cable materials and their functions and characteristics
Learning Outcome 3:	KT0403 Identification of cable characteristics and properties
Learning Outcome 4:	KT0404 Applications of various types of cable
Learning Outcome 5:	KT0405 Installation methods and safe use of cables
Learning Outcome 6:	KT0406 Safe transport and storage of cables

Task	Questions Description	Marks
4	a) Describe, with the aid of sketches, the different types of cables and their construction, and explain the function of the various materials used	4
	b) Describe the factors effecting the effeciency of cables	3
	c) Describe the precautions required when transporting and storing cables	3



Electrical cables consist of conductors (copper/aluminium) for current transmission, insulation (PVC/XLPE) to prevent short circuits, and a protective sheath (PVC/rubber). Key types include LV/MV power cables, armoured cables (SWA/AWA) for mechanical protection, and flexible, flat, or fiber-optic data cables.

(c) Conductor material and conductivity; Copper and aluminum are standard due to low resistivity; higher purity materials offer better conductivity.

Cross-sectional area (Thickness)! Thicker cables have lower resistance and allow more current to flow with less power loss, increasing efficiency.

Cable length! Longer cables have higher total resistance, leading to increased voltage drop and lower efficiency.

Temperature! Higher ambient temperatures increase conductor resistance, reducing efficiency.

(d) Properly transporting and storing cables requires keeps them dry, sealing cable ends to prevent moisture ingress, storing them off the ground in cool shaded areas, and preventing mechanical damage (e.g. dropping or harsh bending).

Avoid contact with corrosive materials, store in dry areas, and never drop or throw cable drums.

Loading and unloading: Never throw or drop cable drums from trucks, particularly in cold weather, to avoid damaging the insulation and sheath.

Lifting procedures: When using a crane or forklift, lift only one drum at a time. Ensure forklift forks are properly positioned to avoid damaging the drum or cable.

Coil handling! Do not hoist multiple coils of cables simultaneously.

Storing position! Always store cable drums standing up (rolling-wise) rather than flat. Placing them flat can cause the edge of the flange to crush the cable.

Separation! Store away from acidic, alkaline or corrosive substances.

LEARNING UNIT 5:

KM-07-KT05: Switchgear and control gear (20%)

Learning Outcome 1:	KT0501 Principles of operation of switchgear and control gear
Learning Outcome 2:	KT0502 Components of switchgear and control gear systems and the application thereof.
Learning Outcome 3:	KT0503 Electrical drawings
Learning Outcome 4:	KT0504 Introduction to Programmable Logic Controllers (PLC's)
Learning Outcome 5:	KT0505 Introduction to Soft starters
Learning Outcome 6:	KT0506 Introduction to Variable speed drives

Task	Questions Description	Marks
5	a) Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and operating mechanisms	4
	b) Describe the basic principles of operation for programmable logic controllers	3
	c) Describe the basic principles of operation for variable speed drives	3

Disconnectors (Isolators)

Construction: Comprise robust metal contacts (often "knife" style) mounted on insulators, connected to a manual lever or mechanical operating mechanism for high-voltage and high-current safety isolation.

Operating principle: A mechanical action opens the contacts to create a visible break in the circuit, ensuring safe, de-energized conditions for maintenance. They lack arc quenching mechanisms, meaning they are for off-load switching only.

Contacts: Large, visible, normally-open gap in the open position.

Relays:

Construction: Comprise an electromagnet coil, a movable armature (operating mechanism), and contact sets (normally open/closed).

Operating principle: A low-power control signal energizes the coil, generating a magnetic field that attracts the armature, mechanically moving the contacts into their alternative state (NO close, NC opens).

Operating Coil: Low-voltage/low-current, create a magnetic field when energized.

Contactors

Construction: Similar to relays, but with larger contacts for high-load handling, sturdy arc chutes (arc extinguishing mechanism), and a strong electromagnetic coil, armature, and spring mechanisms.

Operating Principle: 'Electromagnetic switching'; the energized coil pulls the armature, moving the main contacts to close the circuit, allowing heavy loads to run.

Contacts: Large contacts designed for high-current loads.

Timers

Construction: Integrate a timing mechanism (electronic, digital, or pneumatic) into the structure of a relay.

Operating principle: They trigger contact state changes (opening or closing) after a pre-defined time interval.

Timer types: Include on-delay (contacts change state after the coil is energized for a set time) or Off-delay (contacts change state after the coil is de-energized for a set time).

(b) A programmable logic controller (PLC) operates by continuously cycling through a three-step process:

3 reading input signals from sensors, executing user-programmed logic, and updating output signal to actuators. This scan cycle occurs in milliseconds, ensuring fast, reliable industrial automation, including input diagnostics and application-specific commands.

(c) A Variable frequency drive (VFD) controls AC motor speed by converting fixed-frequency, fixed-voltage AC power into variable-frequency, variable-voltage AC power. It operates in three main stages:

3 rectification (AC to DC), DC bus filtering (smoothing DC), and inversion (DC to variable AC), using pulse width modulation (PWM) to dictate motor speed and torque.

LEARNING UNIT 6:

KM-07-KT06: Lighting systems (10%)

Learning Outcome 1:	KT0601 Principles of Illumination
Learning Outcome 2:	KT0602 Types of luminaries and lighting systems

Task	Questions Description	Marks
6	a) List the different types of alternating current motors and describe their construction	2
	b) Describe Characteristics of single phase and three phase motors	4
	c) Describe and explain how tests are conducted on alternating current motors	2
	d) Calculate alternating current motor properties	2

② Alternating current (AC) motors are classified into two primary categories based on speed: Synchronous and asynchronous (induction) motors. They consist of a stationary stator (generating a rotating magnetic field) and a rotor (rotating part). Key types include squirrel-cage induction, wound-rotor induction, and synchronous motors, often customized for single or three-phase power.

② Single phase motor

Starting: Not self-starting, require auxiliary components like capacitors or centrifugal switches.

Power/Speed: Lower power (Typically $< 5\text{KW}$), with lower efficiency and starting torque.

Design: Simpler design, often utilizing two windings (main and auxiliary).

4

Applications: Residential appliances, fans, small pumps and light commercial tools.

Three-phase motor

Starting: Self-starting due to a natural rotating magnetic field generated by 120° spaced windings.

Power/speed: Higher efficiency (10-20% higher) and higher power output.

Design: Robust construction, usually using three windings.

Applications: Industrial machinery, large air compressors, conveyor systems and industrial pumps.

(c) AC motor testing involves a combination of offline static tests (insulation, winding resistance) and online dynamic tests (vibration, load tests) to assess electrical and mechanical integrity. Key methods include using a megohmmeter (megger) to verify insulation resistance ($>100\text{ M}\Omega$), multimeters to check winding balance (within 1%-2%) and surge testing for turn-to-turn faults.

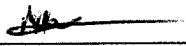
(d) Example: A 4pole motor at 60Hz has a synchronous speed of $= \frac{120 \times f}{P} = \frac{120 \times 60}{4} = 1800\text{ RPM}$

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	KT0101 Theories and concepts of alternating current			✓
2	KT0102 Fundamental principles of alternating current			✓
3	KT0103 Alternating current generation (distribution systems theory)			✓
4	KT0601 Principles of Illumination			✓
5	KT0602 Types of luminaries and lighting systems			✓


Learner Signature

Date


Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT

Learner's Total Mark:

Requirements met ☐

Requirements not met ☐

Action/s required:

By when:

LEARNER FEEDBACK & COMMENTS

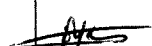
Good work

DECLARATION BY THE FACILITATOR

I (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER

I, Munyisi Mdalabeni.....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.


Learner**Date**
Facilitator**Date****DECLARATION BY THE ASSESSOR**

I, Munyisi Mdalabeni..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.


Assessor**Date****Moderator****Date**

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETPD Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire

2. Observation : As per Practical Assessment Guide

m

3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments
4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit

M. Ndabeni

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Module #	671101000-KM-07:
NQF Level	level 4,
Notional hours	310
Credit(s)	Cr 31
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

SUMMATIVE ASSESMENT

Details	Please Complete this Section
Name	Mvuyisi
Surname	Ndabeni
Organisation:	EASTC TECHNOCENTRIC VARSITY
Unit/Dept:	ELECTRICAL ENGINEERING
Facilitator Name:	

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	Module TITLE	NQF level	Credits	✓
671101000-KM-07	ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS	4	31	

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Muyisi Mdabeni</i>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Muyisi Mdabeni*

Learners Name: *Muyisi Mdabeni*Learner's signature: *[Signature]*

Date: _____

Assessors Name: _____

Assessor's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Munyizi Ndabani</i>	Signature: <i>[Signature]</i>
Assessors Name:	Signature:



INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:
KM-07-KT01: Concepts, theories and principles of supply Systems
(20%)

Learning Outcome 1:	KT0101 Theories and concepts of alternating current
Learning Outcome 2:	KT0102 Fundamental principles of alternating current
Learning Outcome 3:	KT0103 Alternating current generation (distribution systems theory)
Learning Outcome 4:	KT0104 Characteristics and calculations from alternating current waveforms
Learning Outcome 5:	KT0105 Theories, concepts and principles of direct current sources
Learning Outcome 6:	KT0106 Sources of direct current
Learning Outcome 7:	KT0107 Calculations of direct and alternating current circuits
Learning Outcome 8:	KT0108 Construction and operating principles of direct current generators
Learning Outcome 9:	KT0109 Characteristics, sources and generation of renewable energy
Learning Outcome 10:	KT0110 Relevant legislative requirements

Task	Questions Description	Marks
1	a) Explain the principles of generation by using Fleming's right hand rule and Faraday's laws	4
	b) Explain the load balancing principles in a three phase supply system.	4
	c) Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account	4
	d) Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods.	4
	e) Describe with the aid of drawings the components of direct current generators and their functions.	4

(3) When a conductor such as a wire attached to a circuit moves through a magnetic field, an electric current is induced in the wire due to Faraday's law of induction. The current in the wire can have two possible directions. Fleming's right-hand rule gives which direction the current flows.

(14) Load balancing in a three-phase system ensures distribution of electrical loads ($P_A = P_B = P_C$) across all three phases to maintain stable voltages, minimize neutral current to near zero, and prevent equipment overheating. It maximizes efficiency, extends equipment life and prevents tripped breakers by keeping phase impedances equal, ensuring current flow is identical across all phases.

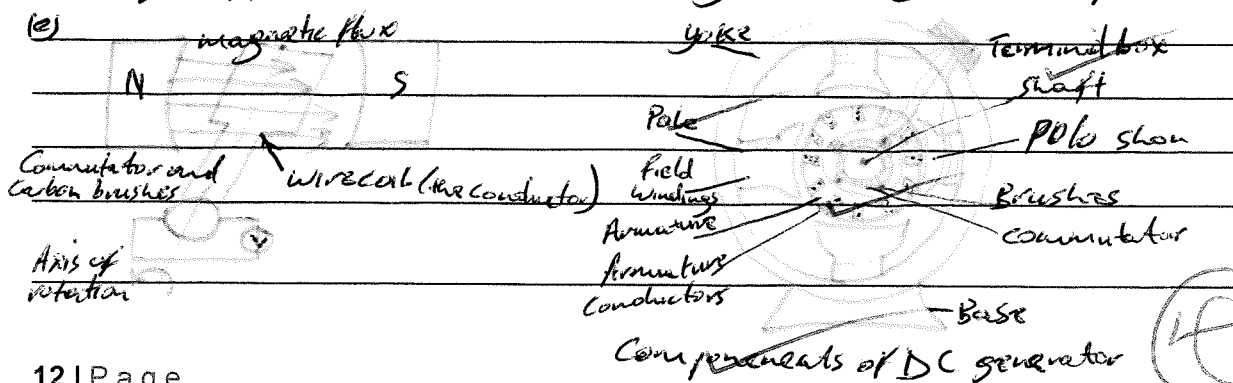
(c) Give a 415V line voltage 25Ω resistance and negligible reactance in a delta load:

$$V_{ph} : 415V$$

$$I_{ph} = \frac{V}{R} = \frac{415V}{25\Omega} = 16.6A$$

$$I_L = I_{ph} \times \sqrt{3} = 16.6 \times \sqrt{3} = 28.75A$$

(d) Renewable energy is generated from naturally replenishing sources like sunlight, wind, and water, offering clean alternative to fossil fuels. Key methods include solar, wind, hydropower, geothermal, and biomass, which reduce carbon emissions. While beneficial for air quality and sustainability, these methods often face challenges like high upfront costs, intermittency and large land requirements.



LEARNING UNIT 2:

Batteries

KM-07-KT02: (10%)

Learning Outcome 1:	KT0201 Fundamentals of battery operation
Learning Outcome 2:	KT0202 Types and classification of batteries
Learning Outcome 3:	KT0203 Safety precautions of batteries
Learning Outcome 4:	KT0204 Care and maintenance of batteries
Learning Outcome 5:	KT0205 Battery components and construction
Learning Outcome 6:	KT0206 Disposal of batteries

Task	Questions Description	Marks
2	a) Explain the terms potential difference, electromotive force, relative density and capacity	3
	b) Describe and explain the various types of batteries and cells	3
	c) Describe and explain the correct procedures to care, maintain, store and dispose of batteries and the hazards and safety precautions associated with batteries and battery rooms	4

Potential difference (Voltage): The energy lost or work done by a unit charge passing between two points in a circuit, measured in Volts (V).

Electromotive force (EMF): The maximum electrical energy per unit charge supplied by a source (battery/generator) when no current is flowing (open circuit).

Relative density: The ratio of the density of a substance to the density of a reference substance. It is dimensionless.

Capacity (battery capacity): The total amount of electrical energy a battery can store and deliver, usually measured in Amperes-hours (Ah) or milliampere (mAh).

b) Batteries and cell are electrochemical devices that convert stored chemical energy into electrical energy, categorized primarily into non-rechargeable (primary) and rechargeable (secondary) types. Primary cells (non-rechargeable)

3 Alkaline Batteries: Utilize an alkaline electrolyte (potassium hydroxide), offering a long shelf life and high power for household items like remote controls, toys and flashlights.

Secondary cells (rechargeable):

Lithium-ion (Li-ion) batteries: Known for their light weight, high energy density, and lack of memory effect. Used in laptops, smartphones and electric vehicles (EVs).

Lead-acid batteries: Known for their capability to provide high surge current and low cost, used in automobiles and UPS systems.

k) Proper battery management is essential to maximize lifespan, ensure safety, and prevent environmental hazards. Key procedures involve storing batteries in cool, dry place, keeping them clean, ensuring proper charging with compatible equipment and disposing of them through specialized recycling channels rather than household trash.

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<u>LEARNING UNIT 3:</u> KM-07-KT03: Transformers (20%)

Learning Outcome 1:	KT0301 Theories of single phase wound transformers
Learning Outcome 2:	KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.
Learning Outcome 3:	KT0303 Fundamentals of transformer construction
Learning Outcome 4:	KT0704 Transformer cooling systems
Learning Outcome 5:	KT0305 Principles of single phase transformer operation
Learning Outcome 6:	KT0306 Principles of single phase auto-transformer operation
Learning Outcome 7:	KT0307 Transformer losses
Learning Outcome 8:	KT0308 Formulas and calculations on input and output of transformers
Learning Outcome 9:	KT0709 Maintenance requirements of transformers
Learning Outcome 10:	KT0310 Types of three phase transformers and their applications
Learning Outcome 11:	KT0311 Construction of three phase transformers including open core, closed core and shell or divided core
Learning Outcome 12:	KT0312 Principles of 3 phase transformer operation and configuration
Learning Outcome 13:	KT0313 Principles of 3 phase auto transformer operation and configuration
Learning Outcome 14:	KT0314 Transformer protection and phasing

Task	Questions Description	Marks
3	a) Name and describe the types of single phase transformers	3
	b) Describe and explain the cooling systems used on transformers	4
	c) Explain, with the aid of drawings, the principle of operation of single phase auto- transformers	4

d) Describe and explain the maintenance requirements for single/three phase transformers	3
e) Describe, with the aid of diagrams, the construction of three phase transformers	4
f) Calculate the relationship of the voltage, current, turns ratio's and power including true power, apparent power and power factor.	3
g) Describe, with the aid of diagrams, three phase transformer connections	4
h) Calculate the line and phase voltages and currents relating to three-phase transformer connections	6
i) Explain transformer protection and phasing	3

a) Based on Voltage adjustment

Step-up transformer: Increases the secondary voltage by having more windings in the secondary coil compared to the primary coil. Used in specific industrial processes.

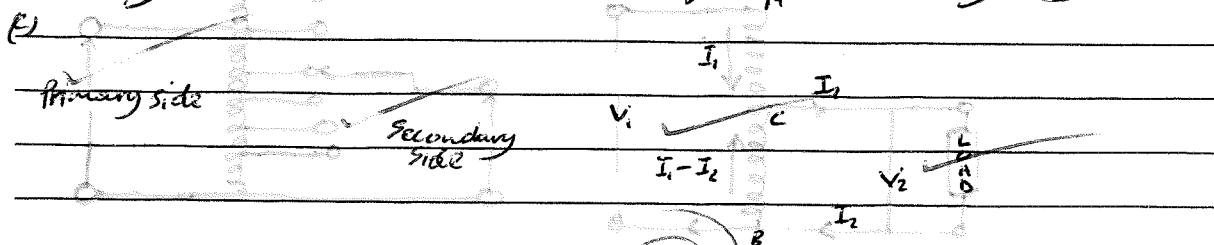
Step-down transformer: Decreases the voltage from the primary to the secondary side, commonly used for lowering power lines to residential voltages.

Based on Construction

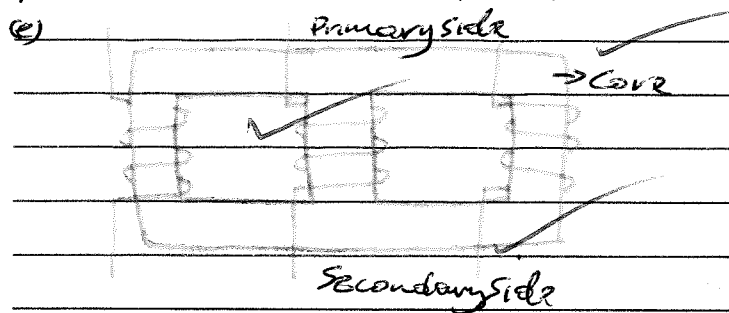
Core-type transformer: Windings surround the core, typically designed as a single magnetic circuit.

(b) Transformer cooling systems dissipate heat generated in windings and cores to prevent insulation degradation, using natural or forced circulation of air, oil or water.

Common methods include ONAN (oil natural, air natural), ONAF (oil natural, air forced) and OFAF (oil forced, air forced) for large power units, crucial for operational longevity.

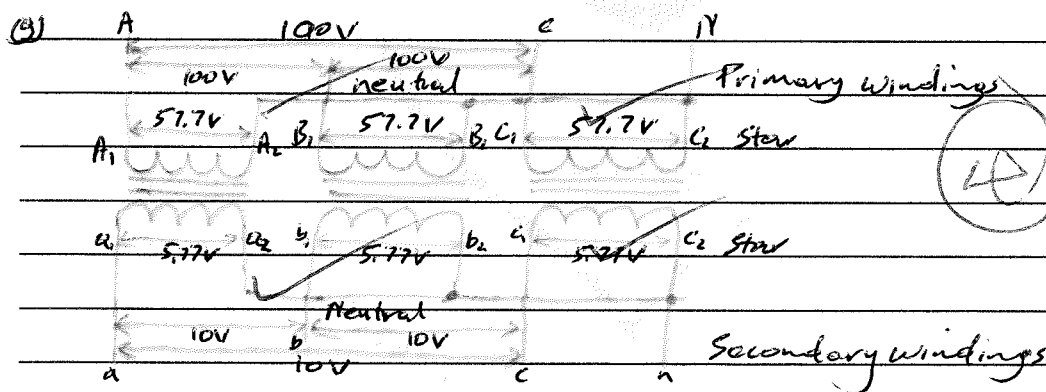


(d) Transformer maintenance involves routine inspections, oil analysis, and electrical testing to ensure reliability and longevity, typically performed annually. Key requirements include cleaning insulators, checking oil level/leaks, testing dielectric strength, verifying bushings and monitoring for hot spots via infrared scans to prevent failures and ensure efficient cooling.



(f) $V_p = 240V$ A transformer has a primary voltage V_p of $240V$ and a turns ratio (N_1/N_2) of $2:1$ $P.F. = 0.9$
 $\frac{V_p}{N_1} = \frac{V_s}{N_2} \Rightarrow \frac{240}{2} = \frac{V_s}{1} \Rightarrow V_s = 120V$ $I_s = I_p \times 2 = 5 \times 2 = 10A$

$P = \frac{V \times I}{P.F.} = \frac{240 \times 5}{0.9} = 1200VA$ $P = S \times P.F. = 1200 \times 0.9 = 1080W$



(h) Star connection: $V_L = \sqrt{3} V_p$ $I_L = I_p$

Delta connection: $V_L = V_p$ $I_L = \sqrt{3} I_p$

If the line voltage is $440V$ & its a delta connection, the V_p is also $440V$

If the line V is $400V$, the phase voltage in star connection across is $\frac{400}{\sqrt{3}} = 230.94V$

i) Transformer protection and phasing ensure power system reliability by guarding against faults (overcurrent, thermal, internal faults) and ensuring correct, synchronized operation of parallel units. Protection involves devices like differential relays and Buchholz relays to isolate faults, while phasing aligns transformer voltage, polarity and vector.

LEARNING UNIT 4:

KM-07-KT04: Types of cables and applications (20%)

Learning Outcome 1:	KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.
Learning Outcome 2:	KT0402 Cable materials and their functions and characteristics
Learning Outcome 3:	KT0403 Identification of cable characteristics and properties
Learning Outcome 4:	KT0404 Applications of various types of cable
Learning Outcome 5:	KT0405 Installation methods and safe use of cables
Learning Outcome 6:	KT0406 Safe transport and storage of cables

Task	Questions Description	Marks
	a) Describe the characteristics and applications of the various types of cables and compare their advantages and disadvantages.	5
4	b) Describe and explain the provisions in the SANS 10142-1 Code of Practice for the correct and safe installation and use of the various types of cables.	5

a) Characteristics and applications

Twisted pair (UTP/STP): Pairs of insulated wires twisted together. Ideal for LAN networking due to affordability and ease of installation.

Coaxial cable: features a central conductor with shielding, designed for RF signals. Commonly used in CCTV, broadband internet and television.

Ribbon cables: Multiple flat conductors used inside computers and electronics to connect components.

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Advantages

Twisted pair: Low cost, flexible, easy installation.

Coaxial: Excellent shielding, higher bandwidth than twisted pair, durable.

Armored: Excellent mechanical protection, safe for hazardous environments.

Disadvantages

Twisted pair: Limited speed/distance compared to fiber; susceptible to interference.

Coaxial: Thicker and less flexible than UTP; more expensive than twisted pair.

Armored: Heavy, rigid and costly.

⑥ Material and construction: Wireways must be made from non-combustible materials and be corrosion-resistant. They must be designed to prevent the ingress of dust and moisture.

Installation & support: All wireways must be securely fixed to prevent movement and supported at intervals not exceeding 1.5m to maintain structural integrity.

Accessibility: Wireways must be installed in locations that allow for maintenance, inspection, and future repairs.

Overcrowding control (cable capacity): Cables must not exceed the allowed percentage fill to prevent overheating.

SANS 10142-1 specifies using a calculated "k" values to determine the correct conduit diameter.

Earthing and bonding: All metal wireways, such as conduit or tray, must be adequately earthed to safely handle potential fault currents.

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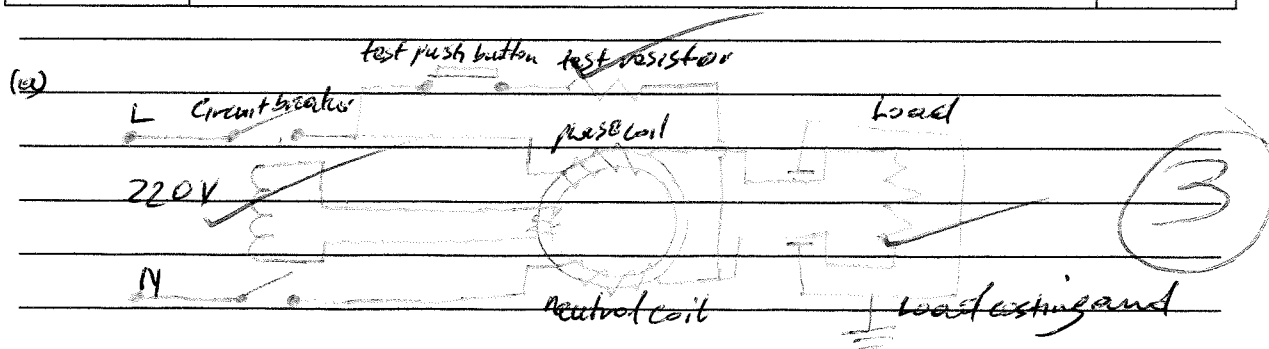
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LEARNING UNIT 5:

KM-07-KT05: Switchgear and control gear (20%)

Learning Outcome 1:	KT0501 Principles of operation of switchgear and control gear
Learning Outcome 2:	KT0502 Components of switchgear and control gear systems and the application thereof.
Learning Outcome 3:	KT0503 Electrical drawings
Learning Outcome 4:	KT0504 Introduction to Programmable Logic Controllers (PLC's)
Learning Outcome 5:	KT0505 Introduction to Soft starters
Learning Outcome 6:	KT0506 Introduction to Variable speed drives

Task	Questions Description	Marks
5	a) Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays	3
	b) Describe the following terminology in terms of low voltage circuit breakers: moulded cases, positive indication, trip position, factory sealed, thermal magnetic tripping, quick make, quick break, trip free mechanism, interpole barriers.	4
	c) Describe the basic principles of operation for soft starters	3



(b) Moulded cases: The outer casing of an MCCB, constructed from high-strength, glass-polyester, or thermoset composite resin.

Positive indication: A safety feature where the physical handle position (on, off or tripped) corresponds exactly to the actual position of the internal contacts.

Trip position: A distinct midway position of the operating toggle handle, located between the "on" and "off" positions, that indicates the breaker has automatically tripped due to a fault (overload or short circuit).

Factory sealed: Indicates that the internal mechanism and trip unit are enclosed within the molded case at the factory, typically to prevent tampering or unauthorized adjustment of trip settings, particularly in UL 489 listed breakers.

Thermal magnetic tripping: A method using two mechanisms for protection.

Quick break: A feature where the opening speed of the contacts is independent of the handle speed.

(c) A soft start works by temporarily reducing the input voltage to an electric motor during startup, limiting inrush current and torque to prevent sudden jerks, mechanical wear, and power surges. Soft starters only manage the start/stop ramp and do not control the motor's speed while it is running.

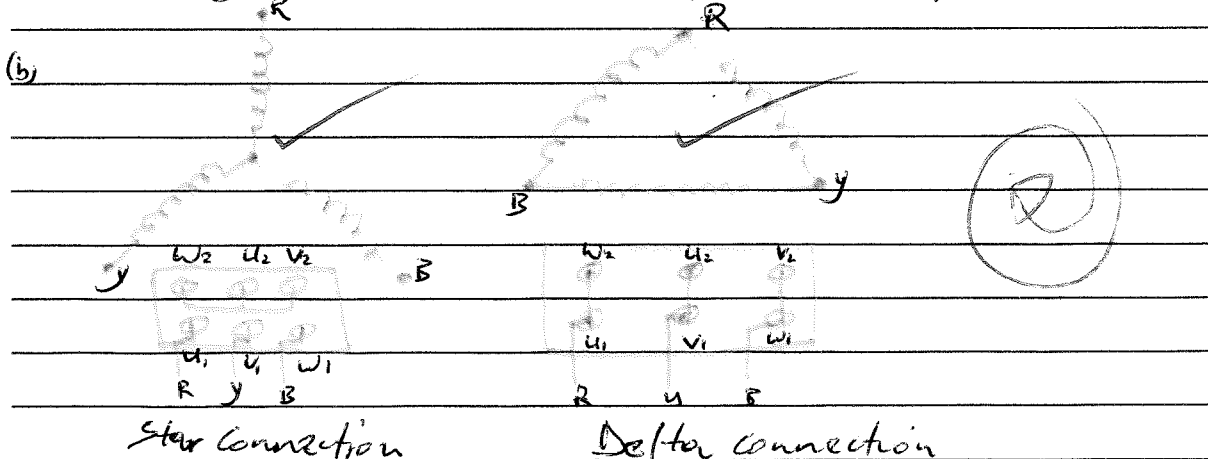
LEARNING UNIT 6:

KM-07-KT06: Lighting systems (10%)

Learning Outcome 1:	KT0601 Principles of Illumination
Learning Outcome 2:	KT0602 Types of luminaries and lighting systems

Task	Questions Description	Marks
6	a) Explain the principle of operation of alternating current motors	2
	b) Describe with the aid of diagrams, the configuration of motor connections	2
	c) Describe and explain protection devices used for alternating current motors	2
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	4
	e) Compare the advantages and disadvantages of single- and three-phase motors	4

(a) AC motors convert alternating current electrical energy into mechanical energy, primarily using electromagnetic induction to produce torque. (2)



(c) Overload relays (thermal/electronic) to stop prolonged overcurrent ~~these~~ fuses or circuit breakers (MCB) for short-circuit protection, phase loss monitors, and thermal sensors (PTC/RTD) for temperature management. (2)

d) Overload protection devices use current ratings to define permissible operating levels and time delays to differentiate between temporary inrush currents and harmful sustained overloads. They prevent motor burnout by tripping based on a motor's heating curve, allowing brief high currents during startup while quickly isolating sustained overloads, locked rotors, or short circuits. (4)

(e) Single-phase motors

Advantages

Cost effective: Generally cheaper to buy and maintain.

Easy Availability: Runs on standard single-phase power, readily available in homes and small shops.

Disadvantages

Not self-starting: Requires a starting mechanism (like a capacitor).

Lower efficiency: Lower power factor and lower energy efficiency compared to 3-phase. (4)

Three-phase motors

Advantages

High efficiency: 10-20% lower energy consumption than single phase.

Self-starting: Starts without auxiliary circuits.

High torque: Produces smooth, constant torque and runs quieter.

Disadvantages

Higher initial cost: More expensive, especially for smaller power needs.

Complex installation: More complex wiring and starting equipment.

Assessor's Report

Candidate's Name	Muayis Ndabeni	ID No.	950115587089
Assessor's Name	Baerinah Tyabedi	Reg. No.	NAMB-A/ 2025/5307

ASSESSMENT DECISION

	C	NYC	Comments

OVERALL COMPETENCE JUDGEMENT:

Comments

Date

ASSESSMENT DECISION			
Source of Evidence	C	NYC	Comment
Knowledge questions	✓		<i>Good</i>
Product /Work sample	✓		<i>Good</i>
Project	✓		<i>Good</i>
Indirect Evidence	✓		<i>Good</i>
Overall Assessment Decision			
Additional Notes			
Date			
			
Signature of Assessor			Signature of Candidate

Candidate Appeal Form

Candidate's Name: <i>Munyisi Ndabeni</i>	ID No.	950115587089
Assessor's Name:	Reg. No.	
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name	Baierinab T	Reg. No.	HAMB-A/2025/5307
Candidate's Name	Mwajiri Mbatia	ID No.	950115587089
MODERATION DECISION			
	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

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ASSESSMENT REVIEW

NAME of LEARNER <i>Murayiz</i>		NAME of ASSESSOR <i>Buainah Jabadi</i>	
VENUE <i>EASTC</i>		DATE of REVIEW	

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☒ Agree ☐ Disagree	☒ Agree ☐ Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
			
Learner	Assessor	Moderator	Date

